

Vehicle Architecture-Driven Design and Comprehensive Loss Analysis of an 800V Three-Level Flying Capacitor Traction Inverter

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Abstract—The design of electric vehicle (EV) traction inverters requires balancing efficiency, torque quality, and semiconductor thermal stress across highly variable driving conditions. This paper presents the specification, simulation, and comprehensive loss analysis of a three-level traction inverter driving a three-phase load. To address modern EV performance demands, an 800V DC bus architecture was explicitly chosen for this design to reduce phase currents and ohmic losses. The selection of the Flying Capacitor (FC) topology over the Neutral Point Clamped (NPC) alternative is detailed, highlighting advantages in thermal symmetry and natural balancing. A detailed conduction and switching loss analysis was conducted utilizing the Graovac-Purschel analytical method and datasheet parameters from a commercial 1200V, 450A SiC module. Results show that an adaptive switching frequency schedule yields a 75% reduction in switching losses during high-current urban driving compared to a fixed 20 kHz baseline. Full time-domain simulations validate the design, demonstrating successful natural capacitor balancing, proper pre-charge handling, excellent output waveforms, and safe thermal operation.

I. INTRODUCTION

The traction inverter is a critical component in Electric Vehicle (EV) powertrains, responsible for converting DC battery power into the variable-frequency AC power required to drive the traction motor. A fundamental challenge in traction inverter design is that the required electrical fundamental frequency is not fixed; it is directly proportional to the vehicle's mechanical speed. During city driving, the inverter operates at low fundamental frequencies but must deliver high phase currents for acceleration. Conversely, highway driving requires high fundamental frequencies but lower sustained currents. Consequently, evaluating inverter performance requires modeling the system across a full spectrum of operating conditions rather than a single nominal point.

To address the performance demands of modern EVs, an 800V DC bus architecture was chosen as the baseline for this design project. While traditional EVs utilize 400V systems, adopting an 800V architecture halves the required phase current for a given power level, drastically reducing I^2R ohmic losses in the inverter, motor windings, and cabling [1]. However, this elevated DC link voltage presents significant design challenges. Traditional two-level voltage source inverters (2L-VSI) require 1200V-class semiconductors. While Silicon Carbide (SiC) devices meet this requirement, their fast switching transitions generate severe dv/dt stress, leading to premature motor insulation failure [2]. To mitigate this, a multilevel inverter topology is required to synthesize the output voltage from multiple discrete levels, inherently reducing the voltage step size and improving the harmonic profile [3].

II. MODELING

This section details the topology selection, component modeling, and the vehicle architecture constraints used to define the inverter operating points.

A. Topology Selection: FC vs. NPC

When designing a three-level traction inverter for an 800V EV architecture, the two primary candidates are the Neutral Point Clamped (NPC) and the Flying Capacitor (FC) topologies. A rigorous comparative analysis dictates the selection of the FC topology for this application based on several critical factors unique to EV traction [4].

A major disadvantage of the NPC topology is the unequal distribution of conduction and switching losses among its semiconductor devices. The outer switches in an NPC inverter process different current profiles than the inner switches, leading to asymmetric thermal stress. This complicates cooling system design and degrades long-term reliability [5]. In contrast, the FC topology distributes voltage and current stresses symmetrically across all four switches in a phase leg. Every switch blocks exactly half the DC bus voltage ($V_{dc}/2$), allowing the use of identical power modules throughout the design. Furthermore, the FC topology eliminates the need for the clamping diodes required in the NPC, removing components from the conduction path and thereby reducing overall conduction losses [6].

EV traction drives are characterized by highly dynamic, rapidly fluctuating load profiles. The NPC topology relies on the neutral point of the DC-link capacitor bank, which is notoriously difficult to keep balanced under transient conditions. The FC topology, however, utilizes redundant switching states to naturally balance the flying capacitor voltage. By actively selecting between redundant states based on the phase current direction, the flying capacitor remains balanced at $V_{dc}/2$ regardless of the load power factor.

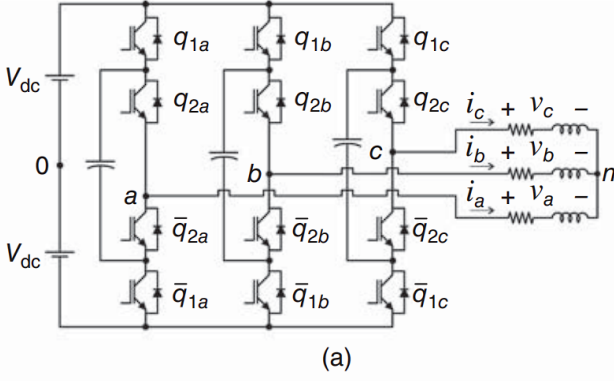


Fig. 1. Three-phase schematic diagram of the 3-level flying capacitor traction inverter. Source: Advanced Power Electronics Converters (E. Cipriano).

B. Component Selection

The Wolfspeed CAB450M12XM3 SiC half-bridge power module was selected for the active switches. The 1200V blocking rating provides a mandatory 50% safety margin over the 800V nominal bus (400V per switch in the 3L-FC configuration) to withstand transient spikes. The 450A continuous current rating provides a robust 3.3x margin over the worst-case peak urban phase current (136A) [7].

The flying capacitor (C_{fc}) must be sized to maintain voltage ripple within a 5% budget (20V on a 400V reference). Using the sizing equation $C = I_{pk} / (2 * f_{sw_min} * \Delta V_{fc})$, with $I_{pk} = 136A$, $f_{sw_min} = 5$ kHz, and $\Delta V_{fc} = 20V$, the required capacitance is 680 μF . Automotive-grade metallized polypropylene film capacitors are specified due to their exceptionally low Equivalent Series Resistance ($ESR < 5$ m Ω) and high ripple current capability [8].

C. Vehicle Architecture and Operating Points

To accurately model inverter losses and performance, the electrical operating points must be mapped from the vehicle's mechanical state. The test vehicle architecture is defined by an 800V DC bus, a single-speed reduction gear (ratio $G = 8.19$), a tyre radius (r) of 0.315 m, and an 8-pole Permanent Magnet Synchronous Motor (pole pairs $p = 4$).

The fundamental electrical frequency (f_e) required by the motor is directly proportional to the vehicle speed (v , in m/s):

$$f_e = (v / r) * (G / 2\pi) * p$$

Urban driving (40 km/h) requires a relatively low electrical frequency of 184 Hz, while highway cruising (120 km/h) demands 552 Hz. The inverter was modeled driving an RL load ($R=1.3\Omega$, $L=2mH$). Because the load impedance increases with frequency, the peak phase current (I_{pk}) is inversely proportional to vehicle speed. Urban driving represents the highest current stress (136 A at 40 km/h), while highway driving draws significantly less current (51 A at 120 km/h).

D. Comprehensive Loss Formulation

To quantify the efficiency impact of the drive frequency, the industry-standard analytical method proposed by Graovac and Purschel [9] was utilized alongside detailed parameters from the CAB450M12XM3 datasheet (R_{ds_on} : 3.3 m Ω at 25°C to 6.6 m Ω at 175°C; $E_{on} = 25.4$ mJ, $E_{off} = 7.51$ mJ at 450A/600V; $R_{th_jc} = 0.094$ °C/W).

Conduction loss is temperature-dependent and scales with the square of the RMS current. The average switching power loss per phase (P_{sw}) for a 3-level inverter can be expressed as:

$$P_{sw} = N_c * (f_{sw} / \pi) * E_{sw_ref} * (I_{pk} / I_{ref}) * (V_{dc_sw} / V_{ref})$$

Where N_c is the number of switches commutating per PWM transition ($N_c = 2$ for 3L-FC).

III. SIMULATION RESULTS

This section presents the results of both the analytical loss optimization and the full time-domain simulation.

A. Analytical Loss and Thermal Optimization

Fig. 2 and Fig. 3 illustrate the conduction and switching losses across the vehicle speed range. Conduction loss is heavily dependent on junction temperature, exhibiting a 2x increase from 25°C to 175°C. Switching loss scales linearly with the carrier frequency (f_{sw}). At a fixed 20 kHz, switching losses dominate at low speeds (urban driving) due to the high phase currents (136A).

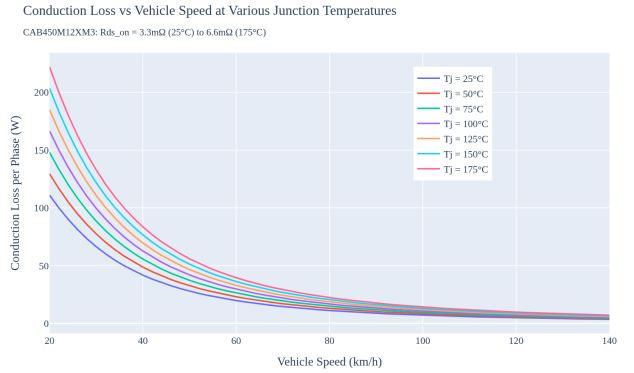


Fig. 2. Conduction loss per phase across the vehicle speed range at various junction temperatures. The loss drops significantly at higher speeds due to the impedance-limited current.

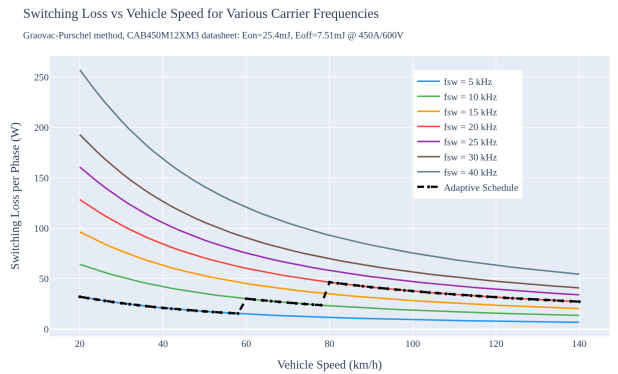


Fig. 3. Switching loss per phase across the vehicle speed range for various carrier frequencies. The dashed line indicates the adaptive switching frequency schedule.

The analytical loss model reveals that operating at a fixed 20 kHz carrier frequency is severely sub-optimal for urban driving. Based on these findings, an adaptive switching frequency schedule was developed and fully implemented within the time-domain simulation. The core methodology of

this adaptive schedule relies on maintaining a sufficient frequency modulation ratio ($m_f = f_{sw}/f_e$) to ensure waveform quality while minimizing losses. During the simulation, as the vehicle operates at low urban speeds, the fundamental frequency (f_e) is low (e.g., 184 Hz at 40 km/h). The controller dynamically reduces the switching frequency to 5 kHz while still maintaining a high modulation ratio ($m_f \approx 27$). This reduction yields massive switching loss savings exactly when the phase current is highest. Conversely, as the simulated vehicle accelerates to highway speeds, f_e increases significantly (e.g., 552 Hz at 120 km/h), and the controller automatically increases the switching frequency to 20 kHz to prevent unacceptable harmonic distortion ($m_f \approx 36$). Because the required phase current is much lower at these speeds, absolute switching losses remain manageable even at 20 kHz. The cycle-by-cycle losses and resulting thermal impacts are extracted directly from this dynamic time-domain simulation.

TABLE I. LOSS AND THERMAL COMPARISON AT DESIGN POINTS

Metric	Urban (40 km/h)	Highway (120 km/h)
Electrical Freq (fe)	183.9 Hz	551.7 Hz
Peak Current (Ipk)	135.8 A	51.0 A
Fixed 20kHz Sw. Loss	84.3 W/phase	31.7 W/phase
Scheduled Sw. Loss	21.1 W/phase (5kHz)	31.7 W/phase (20kHz)
Fixed 20kHz Tj	63.2 °C	46.0 °C
Scheduled Tj	53.5 °C	46.0 °C

As detailed in Table I, the adaptive schedule yields a massive 75% reduction in switching losses during urban driving, equating to a saving of nearly 190 W (3-phase) at 40 km/h. This directly translates to a 9.7°C reduction in semiconductor junction temperature.

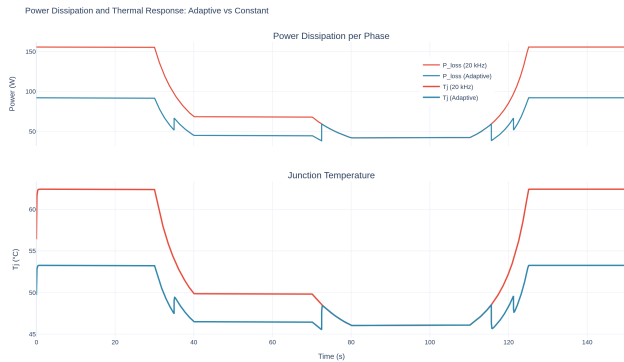


Fig. 4. Power dissipation and transient thermal response over a mixed drive cycle, comparing the adaptive schedule against a fixed 20 kHz baseline.

A transient thermal simulation over a mixed drive cycle (Fig. 4) further demonstrates the benefits of the adaptive schedule. The thermal cycling amplitude is reduced by 53% (from 16.4°C to 7.7°C), which significantly extends the fatigue lifetime of the power modules.

B. Time-Domain Waveforms

To validate the operational stability of the design, a full time-domain simulation was constructed in Python using the datasheet-calibrated parameters. The inverter successfully synthesized the three-level output voltage. Fig. 5 shows the resulting phase current waveforms, and Fig. 6 shows the output voltage waveforms. The Total Harmonic Distortion (THD) of the phase current, calculated using a Hanning-windowed FFT, was excellent: 1.92% for the urban segment (5 kHz) and 1.13% for the highway segment (20 kHz).

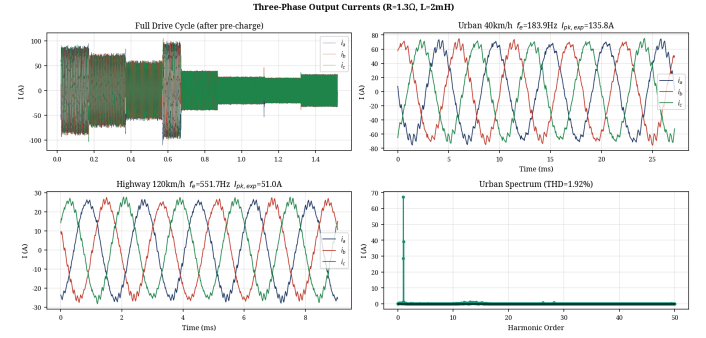


Fig. 5. Simulated three-phase output currents demonstrating proper sinusoidal synthesis.

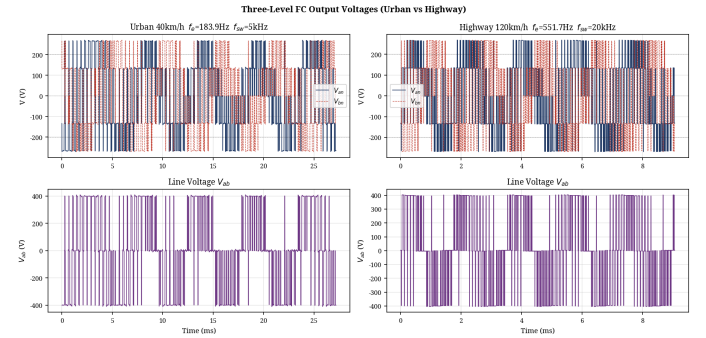


Fig. 6. Three-level output voltage waveforms (line-to-neutral) showing the discrete $V_{dc}/2$ steps.

C. Flying Capacitor Balancing and Pre-Charge

A critical operational requirement is the pre-charging of the 680 μ F flying capacitor to prevent severe overvoltage across the inner switches at startup. The simulation incorporates a 20ms pre-charge sequence, linearly ramping V_{fc} to 400V before active PWM switching commences. As shown in Fig. 7, the PD-PWM balancing algorithm maintained the flying capacitor voltage exceptionally well, with a mean error of less than 0.04V from the 400V reference and a simulated urban ripple of 18.0V (matching the analytical 20V budget).

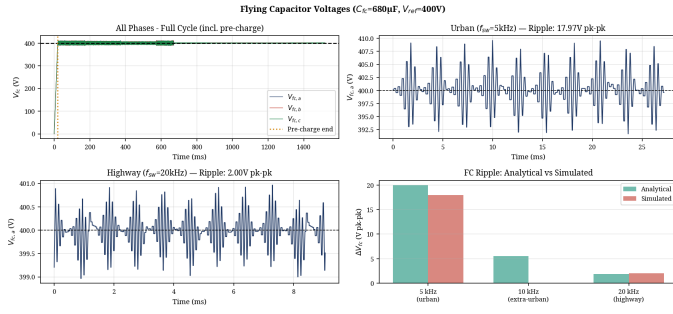


Fig. 7. Flying Capacitor voltage balancing demonstrating the 20ms pre-charge transient and steady-state ripple at urban and highway speeds.

IV. CONCLUSION

This paper demonstrated that traction inverter design and optimization cannot be conducted in isolation from the vehicle architecture. The transition to an 800V architecture provides significant system-level benefits, and the 3L-FC topology was selected over the NPC alternative due to its superior thermal symmetry and natural balancing capabilities under dynamic EV loads. Component selection was rigorously justified using industry standards. By mapping mechanical vehicle speed to inverter electrical parameters, a comprehensive loss profile was generated using the Graovac-Purschel analytical method and CAB450M12XM3 datasheet parameters. Implementing a speed-adaptive switching frequency schedule reduced urban switching losses by 75%, saving 190 W (3-phase) and reducing

thermal cycling amplitude by 53%. The time-domain simulation validated the design, maintaining excellent FC voltage balancing, low THD (<2%), and safe thermal operation. This approach significantly improves overall drivetrain efficiency and reliability where EVs spend the majority of their operational life. It should be noted that the simulation employs ideal switching transitions and does not model parasitic effects such as device voltage drops, discrete dead-time distortion, DC bus ripple, or stray inductance; incorporating these non-idealities in future work would yield more conservative THD and loss estimates.

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